

Module 07: Communication in Robotics

Learning Objectives

1. Define **Communication** within the context of Robotics.
2. Analyze how Communication interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Communication.

Introduction

This module explores **Communication**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Communication is a critical component of the 8-part Active Inference spine, bridging the gap between Learning and Planning.

Key Concepts

1. Communication as a Markov Blanket Boundary

How does Communication define the boundary between the agent and the environment?

2. Generative Models of Communication

What parameters involved in Communication must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Communication drive the perception-action loop?

Applications

In Robotics, we see Communication manifest in: * **Specific Example 1:** A firefly-inspired swarm of aerial robots communicates through synchronized LED flashing patterns, where each robot's generative model predicts the timing and frequency of its neighbors' flashes; phase coupling between oscillators drives the swarm toward synchronization (minimizing collective prediction error), enabling emergent coordination for search-and-rescue operations where RF communication is unreliable -- the robots converge on a shared flashing frequency that signals "area cleared" or "target found" without any centralized message broker. *

Specific Example 2: A bee-waggle-dance-inspired communication protocol allows a foraging robot that has discovered a resource to transmit location information to teammates through a structured motor display (a specific movement pattern encoding distance and direction), which observing robots decode using their own generative models of the dance-to-location mapping; this bio-inspired stigmergic communication reduces bandwidth requirements by orders of magnitude compared to transmitting raw sensor data, while remaining robust to individual robot failures.

Conclusion

Understanding Communication allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.